

After World War II, most cryptography was mathematical in approach, thanks to the easy availability of computers and the discovery of the internet as a communication tool. Shannon was the frontrunner of modern cryptography, relying mostly on mathematics. His extensive work during WWII and publication in the technical journals laid a theoretical base for mathematical cryptography. This was immediately utilized by secret Government organizations like GCHQ and NSA.

In 1970s, the Data Encryption System (DES) made its entry, proposed by a research team at IBM, in a bid to secure electronic communication. This would immensely help business facilities for banks and other financial concerns. After creating a furore, the DES soon became obsolete and was replaced by the Advanced Encryption Standard (AES). Rijndael, a program developed by two Belgian cryptographers were believed to be secure and are still used today. However, this can be highly vulnerable to brute attacks.

Back to the future

The biggest recent development occurred when the Public Key was developed, where a new method of key distribution was introduced. The Diffie-Hellman key exchange created a new class of coding algorithms, using asymmetric keys. Before that, all the keys were rather symmetric in approach, which brought with it a host of problems including the availability of secure channels, especially as the number of participants increase. But the asymmetric key utilizes mathematical keys to decrypt the encryption. By designing a private and public set of keys, the requirement of a secure channel is diminished.

These early ciphers laid the foundation for their modern day counterparts, much more advanced and better suitable for delicate transactions. Right from credit card transactions to internet safety, cryptography is now an integral part of our lives. And its importance has only been multiplied in recent times. 